

Certificate status

Repository source: `certificate/STATUS.md`

Claim

The exact package represents the proposed universal theorem

$$\text{Vol}(K) \geq \frac{26220940089713\pi}{200000000000000} d^3 = 0.41187756378030227424817892598 \dots d^3$$

for every three-dimensional convex body of constant width d .

Current classification

PROPOSED EXACT UNIVERSAL CERTIFICATE — STANDALONE INTERNAL REPLAY

The public-format package is independent of the private repository layout and replays using only files contained in `certificate/`. Its exact Python verifier and independent Python binary64 audit pass in the construction environment. The archived base-R audit passes on the user's machine. Seven adversarial mutations are also rejected by the production verification paths. `Rscript` is not installed in the construction container, so that language path was not rerun there.

Established inside this package

- gap-free circumradius coverage from $1/2$ to `sqrt(6)/4`;
- unique lower-branch bottleneck `03B02`;
- four exact replacement cells for the old `03A1` interval;
- dominant-edge coordinate-width sharpening on all three axes;
- exact final-fan reconstruction and fan-margin verification;
- determinant transfer and forced-core lower bound;
- three exact cap-pair lower bounds and six-cap disjointness;
- strict exact near-Jung handoff surplus;
- exact theorem coefficient and decimal enclosure;
- byte-identical reconstruction of the authoritative certificate JSON;
- independent base-R reconstruction on the user's machine; and
- rejection of targeted lower-row, fan-radius, multiplier, section-hull, cap-allocation, disjointness, and theorem-coefficient mutations;
- automatic reconciliation of the lean Paper v1 theorem constants and lower-branch table against the authoritative certificate JSON; and
- paired Markdown/PDF renderings of the current root, certificate, and paper guides, with PDF preflight and page-by-page manuscript inspection.

Not established

- external independent reproduction;
- peer review or publication;
- sharpness, equality, or rigidity;
- identification of a minimizer;
- Meissner extremality;

Remaining release gates

1. the complete release-document set, including reproducibility, trust-boundary, source-ledger, audit, citation, license, and release notes in paired formats;
2. final authorship, affiliation, license, citation, tag, and release metadata;
3. clean tagged-checkout replay and release-asset verification;
4. external mathematical review or reproduction when available; and
5. user approval before creation of a public remote or announcement.